

**САНКТ-ПЕТЕРБУРГСКИЙ НАЦИОНАЛЬНЫЙ
ИССЛЕДОВАТЕЛЬСКИЙ УНИВЕРСИТЕТ ИТМО**

Дисциплина: Фронт-энд разработка

Отчет

Лабораторная работа 3

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Задача

Мигрировать ранее написанный сайт на фреймворк Vue.JS.

Минимальные требования:

- Должен быть подключён роутер
- Должна быть реализована работа с внешним API
- Разумное деление на компоненты
- Использование composable

Ход работы

Подключение api:

```
import axios from "axios";
import {TokenRef} from "@/composables/tokenActions.ts";

const api = axios.create({
  baseURL: "http://127.0.0.1:8000/",
  timeout: 5000,
  headers: {
    "Content-Type": "application/json",
  },
});

api.interceptors.request.use(
  (config) => {
    const token = TokenRef.value;
    if (token && token !== "undefined") {
      config.headers.Authorization = `Token ${token}`;
    }
    return config;
  },
  (error) => {
    return Promise.reject(error);
  }
);

api.interceptors.response.use(
  (response) => response,
  (error) => {
    console.log(error);
    if (error.response && error.response.status === 401) {
      console.error("Unauthorized!");
    }
  }
);
```

```
        return Promise.reject(error);
    }
});

export default api;
```

Настройка роутера:

```
import { createRouter, createWebHistory } from 'vue-router'
import HomeView from '../views/HomeView.vue'

const router = createRouter({
  history: createWebHistory(import.meta.env.BASE_URL),
  routes: [
    {
      path: '/',
      name: 'home',
      component: HomeView,
    },
    {
      path: "/register", name: "register", component: () =>
import("@/views/Register.vue"),
    },
    {
      path: "/login", name: "login", component: () =>
import("@/views/Login.vue"),
    },
    {
      path: "/workouts", name: "workouts", component: () =>
import("@/views/WorkoutList.vue"),
    },
    {
      path: "/workouts/:id/", name: "workouts details", component: () =>
import("@/views/WorkoutDetails.vue"),
    },
    {
      path: "/workouts/:id/edit",
      name: "EditWorkout",
      component: () => import("@/views/EditWorkout.vue"),
    },
    {
      path: "/workouts/new",
      name: "CreateWorkout",
      component: () => import("@/views/CreateWorkout.vue"),
    },
    {
      path: "/progress",
      name: "ProgressListCreate",
```

```

        component: () => import("@/views/ProgresListCreate.vue"),
    },
    {
        path: '/blogposts',
        name: "blogposts",
        component: () => import("@/views/BlogPostList.vue")
    },
    {path: '/account', name: "account", component: () =>
import('@/views/AccountView.vue')}
    ],
})

export default router

```

Composables:

AccountActions:

```

import {ClearAuthToken, SetAuthToken} from "@/composables/tokenActions.ts";
import api from "@/services/axios.ts";

interface RegisterResponse {
    email: string;
    username: string;
    id: number;
}

interface LoginResponse {
    auth_token: string;
}

interface LoginError {
    username?: string[];
    password?: string[];
    non_field_errors?: string[];
    detail?: string;
}

export interface UserInfo{
    id: 1,
    email: string
    username: string
    first_name: string
    last_name: string
    is_superuser: boolean
}

export function logout() {

```

```

    ClearAuthToken()
  }

export async function login(username: string, password: string): Promise<void>
{
  try {
    const response = await api.post<LoginResponse>('auth/token/login', {
      username: username,
      password: password,
    })

    SetAuthToken(response.data.auth_token)
  } catch (error: any) {
    const errorData = error.response?.data as LoginError | undefined;
    if (errorData) {
      console.log("Login failed", errorData)
    } else {
      console.log("Unexpected error", error)
    }
  }
}

export async function register(email: string,
                                password: string,
                                re_password: string,
                                username: string): Promise<void> {
  const response = await api.post<RegisterResponse>('auth/users/', {
    email: email,
    password: password,
    re_password: re_password,
    username: username
  })
  if (response.status === 201) {
    await login(username, password)
  } else {
    throw new Error("Unexpected error")
  }
}

export async function isAdmin(): Promise<boolean> {
  try {
    const response = await api.get<UserInfo>('fitness/current-user/')
    return response.data.is_superuser
  } catch (e) {
    return false
  }
}

```

ProgressActions:

```
import api from "@/services/axios.ts";

export interface ProgressEntry {
  date: string;
  weight: number;
  notes: string;
}

export async function getProgress(): Promise<ProgressEntry[]> {
  try {
    const response = await api.get<ProgressEntry[]>("/fitness/progress/");
    return response.data;
  } catch (error) {
    console.error("Failed to fetch progress entries:", error);
    return []
  }
}

export async function postProgress (newProgress: ProgressEntry) {
  try {
    const response = await api.post("/fitness/progress/", newProgress);
    if (response.status === 201) {
      alert("Progress entry added successfully!");
    }
  } catch (error) {
    console.error("Failed to add progress entry:", error);
    alert("Failed to add progress entry. Please try again.");
    throw error;
  }
}
```

tokenActions:

```
import {ref, computed} from 'vue';

export const TokenRef = ref(localStorage.getItem('AuthToken'));

export const IsAuthenticated = computed(() => !!TokenRef.value);

export const SetAuthToken = (token: string) => {
  TokenRef.value = token;
  localStorage.setItem('AuthToken', token);
};
```

```
export const ClearAuthToken = () => {
  TokenRef.value = null;
  localStorage.removeItem('AuthToken');
};
```

WorkoutActions:

```
import api from "@services/axios.ts";
import router from "@router";

export interface workoutData {
  id: number;
  title: string;
  description: string;
  video_url: string;
  level: string;
  workout_type: string;
  duration_minutes: number;
  created_at: string;
}

export interface workoutCreateData{
  title: string;
  description: string;
  video_url: string;
  level: string;
  workout_type: string;
  duration_minutes: number;
}

export async function getAllWorkouts(args?: string) {
  try {
    const response = await
api.get<workoutData[]>(`fitness/workouts/?${args}`);
    return response.data;
  } catch (error) {
    console.log(error);
    return []
  }
}

export async function getWorkout(id: string) {
  try {
    const response = await
api.get<workoutData>(`fitness/workouts/${id}/`);
    return response.data;
  } catch (error) {
    console.log(error);
  }
}
```

```

        return undefined
    }
}

export async function editWorkout(workout: workoutData) {
    try {
        const response = await
api.patch<workoutData>(`/fitness/workouts/${workout.id}/`, workout);
        if (response.status === 200) {
            return true;
        }
    } catch (error) {
        console.error("Failed to update workout:", error);
        return false;
    }
}

export async function createWorkout(workout: workoutCreateData) {
    try {
        const response = await api.post<workoutData>("/fitness/workouts/",
workout);
        if (response.status === 201) {
            alert("Workout created successfully!");
            router.push(`/workouts/${response.data.id}/`);
        }
    } catch (error) {
        console.error("Failed to create workout:", error);
        alert("Failed to create workout. Please try again.");
    }
}

```

YouTubeActions:

```

function extractVideoId(videoURL: string): string | undefined {
    if (videoURL.includes('watch')) {
        return videoURL.split('watch?v=').pop()?.split('&')[0]
    }
    return videoURL.split('/').pop()?.split('?')[0]
}

export function convertVideoToThumbnail(videoURL: string) {
    return
`https://img.youtube.com/vi/${extractVideoId(videoURL)}/mqdefault.jpg`
}

```

Вывод

Научились работать с фреймворком Vue.js и настраивать взаимодействие с API.